

LONGSTAFF BLUFF, NU, Canada

WMO: 710910

Lat: 68.899N

Lon: 75.141W

Elev: 161

StdP: 99.41

Time Zone: -5.00 (NAE)

Period: 82-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.8	-39.2	-40.9	0.1	-36.4	-40.3	0.1	-36.3	19.2	-18.6	16.9	-20.3	3.4	90	1.105

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	15.4	10.1	13.7	8.9	12.1	8.0	10.5	14.9	9.2	13.1	8.1	11.3	4.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.5	12.0	6.5	6.1	10.6	5.5	5.7	9.3	30.8	15.2	28.0	13.3	25.3	11.3	13.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.9	12.0	DB	-42.7	17.9	2.6	2.1	-44.6	19.4	-46.1	20.6	-47.5	21.7	-49.4	23.3	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.7	-28.4	-29.4	-26.4	-18.0	-7.4	2.1	8.2	5.9	0.1	-7.7	-16.6	-23.7
	DBStd	14.24	6.81	6.45	6.67	5.99	4.88	3.44	3.36	3.11	3.09	4.85	6.32	7.25
	HDD10.0	7937	1191	1103	1128	840	540	239	77	133	296	548	798	1045
	HDD18.3	10952	1449	1336	1387	1090	798	488	314	385	546	807	1048	1303
	CDD10.0	27	0	0	0	0	0	1	21	6	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.5	5.2	5.1	5.3	5.5	5.6	5.2	4.9	5.3	5.8	6.5	6.5	5.5
Precipitation	PrecAvg	306	13	12	15	19	19	25	36	52	38	37	23	17
	PrecMax	493	37	48	58	68	40	68	93	84	73	84	89	41
	PrecMin	132	1	1	1	3	1	8	11	6	8	7	3	4
	PrecStd	85	9	10	12	15	9	12	17	18	17	18	16	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.2	-8.6	-5.9	-1.5	4.3	13.2	18.2	16.4	8.7	0.8	-1.9	-3.1
		MCWB	-5.8	-9.2	-6.3	-1.9	2.6	8.6	11.3	9.9	5.0	0.0	-2.4	-4.1
	2%	DB	-11.8	-13.4	-10.3	-5.2	1.7	10.7	16.3	14.2	6.8	-0.1	-3.6	-6.9
		MCWB	-12.2	-13.7	-10.6	-6.0	0.8	6.9	10.4	9.5	4.1	-0.2	-4.0	-7.5
	5%	DB	-15.0	-17.0	-14.0	-7.2	0.3	8.9	14.9	12.2	5.4	-0.8	-5.1	-10.6
		MCWB	-15.4	-17.4	-14.3	-7.9	0.0	5.7	9.8	8.7	3.7	-1.0	-5.4	-11.0
	10%	DB	-18.5	-20.4	-16.8	-9.5	-0.6	7.2	13.5	10.6	4.3	-1.8	-7.4	-13.6
		MCWB	-18.9	-20.7	-17.1	-9.9	-0.9	4.7	8.9	7.0	3.0	-2.2	-7.8	-14.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.7	-10.9	-7.2	-5.4	3.1	9.8	12.1	9.1	5.8	0.0	-3.3	-5.2
		MCDB	-6.4	-10.5	-6.9	-4.9	4.3	13.7	17.3	12.1	7.7	0.1	-3.2	-5.0
	2%	WB	-13.8	-14.9	-13.4	-7.0	1.2	7.4	11.2	7.6	4.8	-0.8	-5.0	-9.9
		MCDB	-13.4	-14.7	-13.0	-6.3	2.2	11.0	16.0	9.8	6.1	-0.8	-4.7	-9.6
	5%	WB	-17.0	-18.0	-16.7	-8.5	0.0	6.3	10.4	6.5	3.8	-1.8	-7.6	-13.5
		MCDB	-16.7	-17.6	-16.4	-8.0	0.5	9.5	14.9	8.5	5.1	-1.7	-7.3	-13.1
	10%	WB	-19.7	-21.2	-19.7	-10.6	-1.5	5.1	9.4	5.8	2.9	-2.9	-9.9	-16.0
		MCDB	-19.3	-21.0	-19.5	-10.2	-1.2	7.6	13.4	7.6	4.0	-2.5	-9.5	-15.6
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.5	5.7	5.7	4.3	3.9	5.3	4.4	3.0	3.9	5.4	5.5
		MCDBR	7.7	8.1	7.3	6.4	4.3	6.1	6.7	6.4	4.5	3.5	6.1	6.7
		MCWBR	7.5	8.9	7.0	6.6	3.5	4.0	3.8	3.4	2.9	3.7	8.2	6.2
	5% WB	MCDBR	7.3	8.7	6.7	6.5	3.8	6.0	6.4	4.9	4.1	3.4	7.4	6.1
		MCWBR	7.1	8.4	6.7	6.4	3.5	4.1	4.0	3.3	3.0	3.5	7.4	6.1
Clear-Sky Solar Irradiance	taub	0.158	0.197	0.244	0.299	0.320	0.326	0.341	0.319	0.286	0.231	0.154	N/A	
	taud	1.860	2.072	2.199	2.078	2.126	2.325	2.429	2.528	2.511	2.129	1.908	N/A	
	Ebn at Noon	70	612	800	834	859	870	843	829	765	529	63	0	
	Edh at Noon	5	42	73	114	123	105	92	75	57	39	4	0	
All-Sky Solar Radiation	RadAvg	0.02	0.41	2.06	4.33	5.99	6.21	5.18	3.50	1.83	0.58	0.06	0.00	
	RadStd	0.00	0.06	0.10	0.20	0.39	0.73	0.46	0.30	0.17	0.08	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.59	+1.26	N/A	N/A	N/A	N/A	-240	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.59	+1.26	N/A	N/A	N/A	N/A	-240	N/A	N/A	N/A

Nomenclature: See separate page